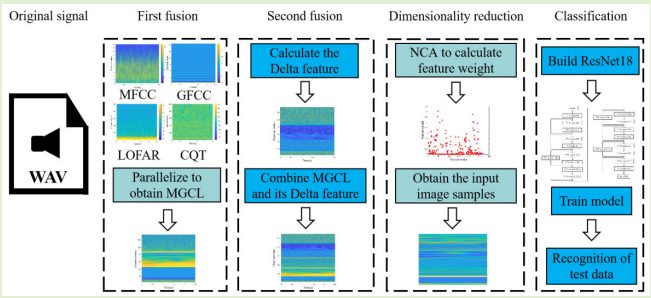
Underwater Acoustic Target Recognition Method Based on Feature Fusion and Residual CNN

Yixin Yang, *Member, IEEE*, Qihai Yao, and Yong Wang, *Member, IEEE*

***Abstract*—This article presents an underwater acoustic target recognition method using feature fusion and residual convolutional neural network (CNN). Mel-frequency cepstrum coefficient (MFCC), Gammatone frequency cepstral coefficient (GFCC), low-frequency analyzer and recorder (LOFAR) spectrum, and constant Q transform (CQT) are extracted and fused first. On this basis, their Delta features are calculated and fused second. The feature dimension is reduced by neighborhood component analysis (NCA). With the fused features after the dimensionality reduction as input features, the residual CNN based on the ResNet18 model is used as classifier to recognize the underwater acoustic target. The other machine-learning models, such as support vector machine (SVM), VGG19, and common CNN, are also compared for inputting different features separately. Experimental results show that, MGCL-Delta-NCA-ResNet18 has the best recognition results among these models, with the recognition accuracy of 97.29%, because this model allows full play to the rich information advantages of feature fusion, advantages of feature dimensionality reduction by the NCA and the ability of ResNet18 to extract abundant characteristics. It can also realize the recognition effectively at low signal-to-noise ratio (SNR). Especially at 0 dB, the recognition accuracy can still reach 86.25%. The proposed method can also recognize multitarget signal effectively in the multiple target scenario. Although this model is used in ship and natural voice recognition, it can also be applied to the recognition of other target sounds, such as marine mammals.**



***Index Terms*— Acoustic target recognition, feature fusion, machine learning, residual convolutional neural network (CNN).**

I. Introduction

UNDERWATER acoustic target recognition, as a forefront technology in ocean exploration, has played a massive role in military and civilian fields and has always been of great concern. Underwater acoustic target recognition has long faced enormous challenges due to the complex and ever-changing marine environment and scarce underwater acoustic data. With the advancement of artificial intelligence technology, applying machine-learning methods to recognize underwater acoustic targets has become a hot spot. It is achieved by processing the acoustic signal, extracting corresponding underwater acoustic features as input, and using the appropriate classifier for recognition.

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The authors are with the School of Marine Science and Technology, Northwestern Polytechnical University, Xi'an 710072, China (e-mail: yxyang@nwpu.edu.cn; 2019260659@mail.nwpu.edu.cn; yongwang@nwpu.edu.cn).

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Several studies have made contributions in the underwater acoustic target recognition field based on the machine learning. Kamal used an unsupervised learning method to extract depth features, most of which were invariant and could be used as discriminant features for recognition [1]. Cao et al. [2] proposed sparse autoencoders to determine the invariant characteristics of spectral data of underwater targets. Chen and Xu [3] established a deep belief network (DBN) and a superimposed denoising automatic encoder (SDAE) model to recognize underwater acoustic targets. Hu et al. [4] used a deep extreme learning machine (ELM) architecture for the recognition of civilian ships. Li et al. [5] achieved the data augmentation by underwater acoustic modeling and used transfer learning to reduce the difference between the augmented data and the measured data, improving the recognition performance and noise resistance of the model. Li et al. [6] proposed multiscale dispersion Lempel–Ziv complexity, which reflects the complexity of ship-radiated noise signals more comprehensively. Wang et al. [7] used a multibranch backbone network coupled with a convolutional attention network, called AMNet, which adaptively selects the effective features by weighting them against the global information of the time-frequency map. Feng et al. [8] proposed the masking hierarchical tokens method, which can learn intrinsic

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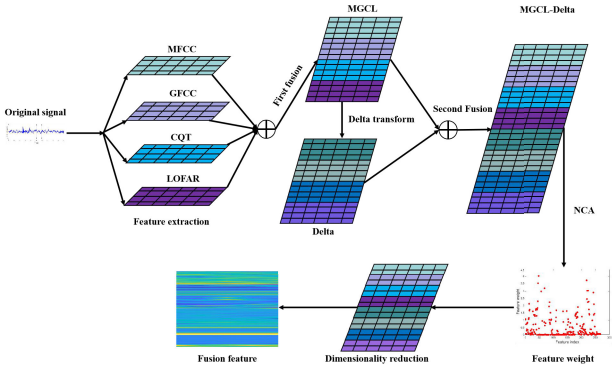


Fig. 1. Framework for the fusion of multiple features.

representations to achieve better recognition performance with fewer labeled data. Cao et al. [9] applied the convolutional neural network (CNN) to attain the temporal correlation of the time–frequency map of signal for the recognition. Wu et al. [10] extracted the LOFAR spectrum of the signal and achieved effective recognition using the CNN. On the basis of using wavelet transform for the denoise of signal, Ghavidel et al. [11] applied few-shot learning to underwater target recognition with only a small number of training samples. Feng et al. [12] trained ResNet18 to obtain output class probabilities with multiple features as the input, and used the weighted average Dempster Shafer (WADS) method for the fusion, significantly improving the recognition accuracy. Ibrahim et al. [13] combined CNN and stacked autoencoder (SAE) to detect whale calls, and the results of this model are remarkably better than those of traditional models. Yang et al. [14] proposed a competitive deep-belief network (CDBN) to extract features with discriminative information.

As for underwater target localization, machine learning has also shown good performance in estimating the range, depth, and azimuth of sound sources. Niu et al. [15] used feedforward neural network (FNN) to estimate the range of the target. Wang and Peng [16] used the generalized regression neural network (GRNN) to estimate the range in shallow water. Ferguson et al. [17] used the cepstrum of the target to achieve the estimation of direction by using CNN. Tang et al. [18] used the CNN for the localization of target with the time-domain signals as input features.

Most of the aforementioned studies recognize based on a single feature. This study presents a recognition method for underwater acoustic targets on the basis of feature fusion technology and residual CNN. Fig. 1 shows the framework for the fusion of multiple features. Mel-frequency cepstrum coefficient (MFCC), Gammatone frequency cepstral coefficient (GFCC), low-frequency analyzer and recorder (LOFAR) spectrum, and constant Q transform (CQT) are extracted and fused first. On this basis, their Delta features are calculated and fused second. The feature dimension is reduced by neighborhood component analysis (NCA). With the fused features after the dimensionality reduction as input features, the residual CNN based on the ResNet18 model is used as classifier to recognize the underwater acoustic target.

II. Underwater Acoustic Feature Extraction Method

To make the machine-learning model recognize the underwater acoustic target effectively, it is necessary to extract feature parameters of the target sound that can be recognized by the model. MFCC, GFCC, CQT, and LOFAR spectrum are extracted in this article.

A. Data Preprocessing

The data preprocessing mainly includes preemphasis, framing, and windowing. Preemphasis enables the frequency spectrum of the signal to be smoother by enhancing the spectrum of the high-frequency part. The signal is divided into several short-term signals by framing, during which the signal can be regarded as a stationary process. In the process of framing, the overlapping segmentation is used to ensure smooth transitions between frames. In order to reduce truncation effects, the original signal $s(n)$ is preprocessed by windowing, resulting in signal $s'(n)$ as follows:

$$s'(n) = s(n)w(n), \quad 0 \le n \le N-1 \tag{1}$$

where N represents the number of samples, and $w(n)$ represents the window function.

B. MFCC

MFCC is presented on the basis of the features of the human ear. Humans can distinguish the low- and high-frequency parts of sound, where the low-frequency part is used to recognize sound features. Dense and sparse filters are set in the low-frequency and high-frequency components to extract effective Mel spectrum, respectively. Through the cepstrum operation, MFCC can be calculated from the Mel spectrum [19]. The extraction process of MFCC is shown in Fig. 2.

On the basis of frame processing, fast Fourier transform (FFT) is used for each frame data to calculate the spectrum. The discrete spectrum $S'(k)$ of the signal is expressed as

$$S'(k) = \sum_{n=0}^{N-1} s'(n)e^{-j2\pi k/N}, \quad 0 \le k \le N. \tag{2}$$

The spectrum is filtered by several triangular bandpass filters to calculate Mel filters. In addition, $f(m)$ represents the center frequency, where $m = 1, 2, \ldots, M$. The triangular filter can be expressed as

$$H_m(k) = \begin{cases} 0, & k < f(m-1) \\ \dfrac{2(k - f(m-1))}{(f(m+1) - f(m-1))(f(m) - f(m-1))}, & f(m-1) \le k \le f(m) \\ \dfrac{2(f(m-1) - k)}{(f(m+1) - f(m-1))(f(m) - f(m-1))}, & f(m) \le k \le f(m+1) \\ 0, & k > f(m+1). \end{cases} \tag{3}$$

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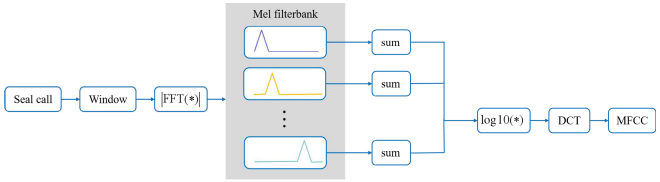


Fig. 2. Extraction process of MFCC.

The logarithmic energy of all filters is obtained as follows:

$$S^*(m) = \ln\left(\sum_{k=0}^{N-1} \left|S'_a(k)\right| H_m(k)\right), \quad 0 \le m \le M. \tag{4}$$

Discrete cosine transform (DCT) is used for the M logarithm energies to calculate the MFCC of order L according to (4), where the DCT can be expressed as

$$C_n = \sum_{m=0}^{N-1} S_m^* \cos\frac{n*(m-0.5)*\pi}{M}, \quad n = 1, 2, \ldots, L. \tag{5}$$

C. GFCC

Gammatone filter refers to a filter based on standard cochlear structure [20], and its time-domain expression is expressed as follows:

$$g_i(t) = At^{n-1}e^{-2\pi b_i t}\cos(2\pi f_i + \phi_i)U(t), \quad t \ge 0, 1 \le i \le N \tag{6}$$

where A is the gain of the filter, f_i is the center frequency of the filter, $U(t)$ is a step function, and ϕ_i is the offset phase. The human ear is not sensitive to the phase offset. Therefore, to simplify the model, this article takes 0. n is the filter order, N is the number of filters, and b_i is the attenuation factor, which will determine the attenuation rate of the current filter to the impulse response. The corresponding relationship with the center frequency is presented as follows:

$$b_i = 1.019 b_{\mathrm{EBR}}(f_i) \tag{7}$$

where, $b_{\mathrm{EBR}}(f_i)$ is the equivalent rectangular bandwidth, and the relationship with the center frequency f_i is as follows:

$$b_{\mathrm{EBR}}(f_i) = 24.7\left(4.37\frac{f_s}{1000} + 1\right). \tag{8}$$

The feature parameter extraction process of GFCC is basically similar to that of MFCC, but the difference is that the Mel triangle filter bank is replaced by the Gammatone filter bank. In this article, n and N are set of 4 and 64, respectively, and the feature parameter G_n is obtained through the above extraction process.

D. CQT

The center frequency of the CQT filter bank exhibits an exponential distribution and has various filtering bandwidths. However, the ratio of center frequency to bandwidth is constant, namely Q [21]. CQT is rich in characteristic information of the low-frequency part of sound signals.

$$X(m,n) = \sum_{i=n-\lfloor N_m/2\rfloor}^{n+\lfloor N_m/2\rfloor} x(i)\left(a_m^*\right)(i-n+N_m/2)$$
$$m = 1, 2, \ldots, M \tag{9}$$

where m represents the number of frequency division points, N_m represents the length of the sliding window, and a_m^* represents complex conjugate of basis function.

E. LOFAR Spectrum

By adding a window to an unstable signal, it is divided into several relatively short signal segments. At short notice, each segment of the signal can be considered a stationary signal, and then subjected to a short-time Fourier transform (STFT):

$$S_x(\omega, t) = \int x(\tau)m(\tau - t)e^{-j\omega t}d\tau \tag{10}$$

where $m(\tau - t)e^{-j\omega t}$ is a window function, which is both a function of angular frequency ω and a function of time t, and can reflect time-varying and periodicity.

F. Delta Transform

Delta feature can extract the trajectory of signals over time, thereby identifying the dynamic change information of signals [22]. Here, the Delta feature is calculated using the information of the previous and following few frames of the current frame, which is expressed as

$$d_t = \frac{\sum_{n=1}^{N} n(c_{t+n} - c_{t-n})}{2\sum_{n=1}^{N} n^2} \tag{11}$$

where d_t represents the Delta coefficient of the tth frame, and the calculation Delta of d_t requires $t - N$ to $t + N$ frames. In this article, N is set to 2. The dimension of final Delta feature obtained is the same as the original dimension.

III. Feature Fusion

MFCC is designed according to the characteristics of human hearing. GFCC is mainly designed according to the frequency induction on the basement membrane, whose parameters can fit strong anti-interference. CQT is rich in low-frequency feature information of sound, and the total amount of feature information in the LOFAR spectrum is relatively large. Therefore, in order to comprehensively improve the richness and noise resistance of feature parameters, this article adopts the above four features and their Delta features as feature fusion parameters. This article first performs the first feature fusion of MFCC, GFCC, CQT, and LOFAR in parallel, called MGCL. The corresponding Delta features of MGCL are also calculated, which particularly contain abundant dynamic information. Then, the first fused features are combined with the Delta features in parallel for the second feature fusion, called MGCL-Delta. The fused features obtained are as follows:

$$\mathrm{MGCL-Delta}_m = \left[(M_1, \ldots, M_m), \ldots, (L_1, \ldots, L_n), \left(M_1^{\mathrm{delta}}, \ldots, M_m^{\mathrm{delta}}\right), \ldots, \left(L_1^{\mathrm{delta}}, \ldots, L_m^{\mathrm{delta}}\right)\right] \tag{12}$$

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where M_m and L_m are MFCC and LOFAR, respectively. M_m^{delta} and L_m^{delta} are the Delta features of MFCC and LOFAR, respectively. GFCC, CQT, and their Delta features are not listed here.

IV. NCA

NCA is a supervised data dimensionality reduction method that maximizes prediction accuracy [23]. Suppose the existing dataset $S = \{(x_i, y_i), i = 1, 2, \ldots, n\}$, where x_i represents a single sample point, the sample dimension is p, and y_i is the corresponding category label. A reference sample point x_i is randomly selected from the dataset, and the category of the next x_j is considered the same as x_i. This consideration is essentially equivalent to a 1-nearest neighbor classifier. The probability that the reference sample point is obtained from the dataset is $P(\text{Ref}(x) = x_j|S)$, and the probability that another point x_j is selected as the reference point is related to the distance of the previous reference point. The distance equation is expressed as

$$d_w(x_i, x_j) = \sum_{r=1}^{p} w_r^2 |x_{ir} - x_{jr}| \tag{13}$$

where w_r is the characteristic weight value. Assume that the probability of being selected is related to the distance of the previous reference point, namely

$$P(\text{Ref}(x) = x_j|S) \propto k(d_w(x, x_j)) \tag{14}$$

where k represents the kernel function for the distance between two points. As $d_w(x, x_j)$ is very few, that is, the distance is very close, the value of k is large and is expressed as follows:

$$k(z) = \exp\left(-\frac{z}{\sigma}\right). \tag{15}$$

Considering that x belongs to dataset S, the sum of $P(\text{Ref}(x) = x_j|S)$ is 1

$$P(\text{Ref}(x) = x_j|S) = \frac{k(d_w(x, x_j))}{\sum_{i=1}^{n} k(d_w(x, x_j))}. \tag{16}$$

Use the method of reserving one to test the above random classifier, that is, reserving test sample (x_i, y_i) and other samples in the dataset as the training set, then the probability that test sample point x_i is selected by the reference sample point of the same category can be expressed as

$$p_{ij} = P(\text{Ref}(x_i) = x_j|S^{-i}) = \frac{k(d_w(x_i, x_j))}{\sum_{j=1,ji}^{N} k(d_w(x_i, x_j))}. \tag{17}$$

The average probability of the random classifier to classify the test samples correctly using the method of leaving one is

$$p_i = \sum_{j=1,ji}^{N} P(\text{Ref}(x_i) = x_j|S^{-i}) l(y_i = y_j). \tag{18}$$

When y_i is equal to y_j, l is 1, otherwise, it is 0. The above equation is simplified as follows:

$$p_i = \sum_{j=1,ji}^{N} p_{ij} y_{ij} \tag{19}$$

where

$$y_{ij} = \begin{cases} 1, & y_i = y_j \\ 0, & \text{else.} \end{cases} \tag{20}$$

Then, the correct classification probability of the random classifier in the whole sample is abbreviated as follows:

$$F(w) = \sum_{i=1}^{N} p_i. \tag{21}$$

Add penalty term to adjust the complexity of the model

$$F(w) = \sum_{i=1}^{N} p_i - \lambda \sum_{r=1}^{p} w_r^2. \tag{22}$$

The importance weight of the feature can be obtained by training the model according to the gradient descent algorithm.

V. CNN

A. Theoretical Basis

The input, hidden, and output layers form the basic framework of the CNN model. The hidden layer contains convolutional, pooling, and fully connected layers [24]. A typical CNN model framework is shown in Fig. 3. Many neurons form the convolutional layer of CNN is derived from the signal transmission mode of neuronal synapses in biology.

In the convolutional layer, neurons are only connected to neurons within the set area of their adjacent convolutional layers. When the data features contain rich local information, this connection method will effectively achieve the training and processing of data. The convolution operation is expressed as follows:

$$f(m, n) * q(m, n) = \sum_{i=-\infty}^{\infty} \sum_{j=-\infty}^{\infty} f(i, j) q(m - i, n - j) \tag{23}$$

where $f(m, n)$ is the feature function and $q(m, n)$ is the convolutional kernel.

The pooling layer realizes the dimensionality reduction of the feature matrix through subsampling, thereby reducing the number of training parameters, which can prevent overfitting and reduce the training time.

The full connection layer summarizes and normalizes the highly abstract feature data processed by the convolution and pooling layers. Based on the activation function, the probability for the classification of each category can be calculated. Finally, the results are obtained by the classification layer.

B. Residual Connection Model

With the increase in the number of network layers, the model can theoretically extract more abundant feature information. However, in the actual training process, the phenomenon of gradient disappearance may occur, leading to the decrease in the recognition performance. Research has shown that the residual connection model can effectively prevent the occurrence of these phenomena [25].

The residual CNN model adopts a skip connection method, as shown in Fig. 4. It can skip some convolutional layers and reduce the training time. It directly transmits important feature

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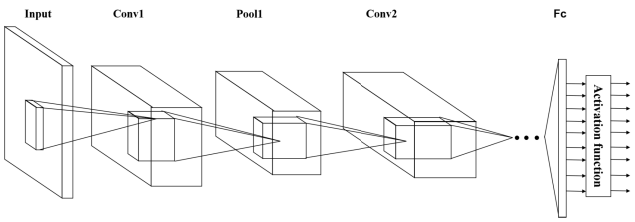


Fig. 3. Typical CNN model.

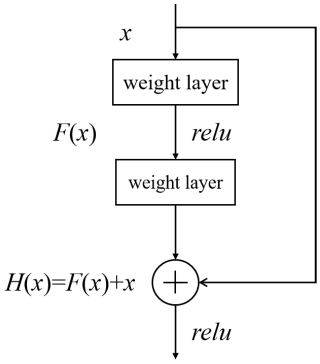


Fig. 4. Structure of the residual connection model.

information to the next layer. The output value of the residual model is

$$H(x) = F(x) + x. \tag{24}$$

C. CNN Model Construction

The ResNet18 model belongs to residual CNN and consists of 17 convolutional layers and one fully connected layer. In this study, the ResNet18 is used as the recognition model, and Fig. 5 shows its architectural diagram. To avoid the pooling layer from filtering out a large amount of feature information, this article removes the pooling layer. The stochastic gradient descent with momentum (SGDM) is used as the algorithm of the optimizer. The batch size is set at 5. The learning rate is set at 0.001.

VI. Simulation Experiment

A. Dataset

The ship audio data used for validation are obtained from the DeepShip database [26], which consists of four types of ships: tug, cargo, tanker, and passenger. The database are collected by a single hydrophone. Table I shows the specific information of ship audios used. The recordings of three ships are selected for each type of ship. Fig. 6 shows the waveforms of the original signals of various targets.

B. Feature Extraction

The audio data are first processed by segments, whose the window length and step size are 256 and 128 samples, respectively. As for MFCC features, 20 groups of filters are set. Therefore, each segment of audio can be calculated to obtain a 1 × 12 vector. In extracting GFCC features, the order of filter is set to 4, and the number of filter is 64. Each audio

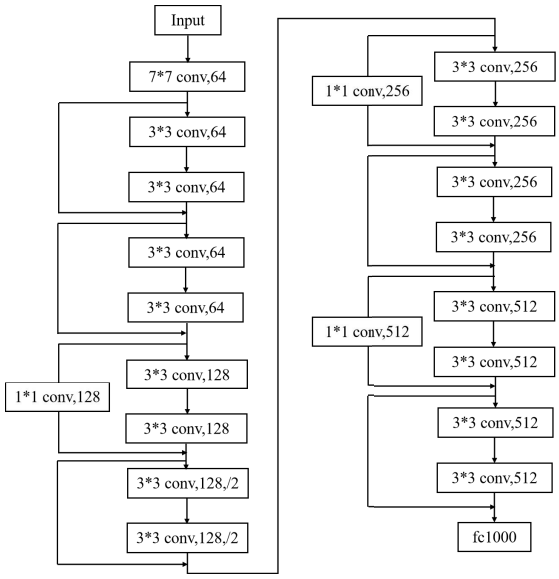


Fig. 5. Framework of ResNet18.

TABLE I
Specific Information of Ship Audios

Category	Specific Number, Name, and Recording Time
Tug	9, MILLENNIUMSTAR, 20171115
	40, SEASPAN COMMANDER, 20171203
	49, SEASPAN EAGLE, 20171210
Cargo	15, NEW NADA, 20171114
	38, ARIES LEADER, 20171123
	62, ANASTASIA, 20171202
Tanker	10, NAVE ORBIT, 20160602
	18, OVERSEAS ATHENS, 20160619
	24, HIGH ENDURANCE, 20160710
Passenger	9, CARNIVAL LEGEND, 20160516
	16, MALASPINA, 20160604
	29, CRYSTAL SERENITY, 20160727

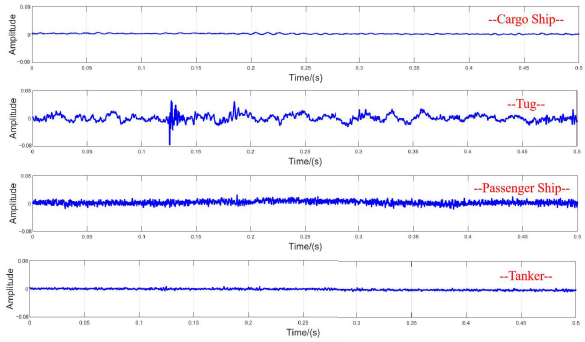


Fig. 6. Waveforms of original signals of different targets.

segment can be calculated to obtain a 1 × 31 vector. For the LOFAR spectrum, the frequency analysis range is set to 0–500 Hz, and the frequency interval is set to 10 Hz. The input feature is the amplitude of each frequency point, whose dimension is 1*51. For the CQT, the number of octave is set to 12. MFCC, GFCC, LOFAR spectrogram, and CQT features of 2.5 s audio data are selected for multiple acoustic targets as an example as shown in Figs. 7–10, respectively.

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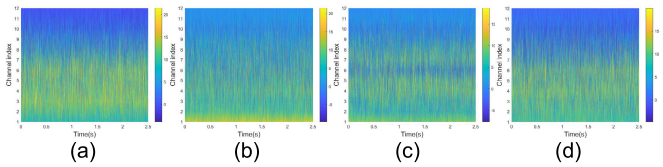


Fig. 7. MFCC characteristics of acoustic signals of different targets. (a) Tug. (b) Cargo. (c) Tanker. (d) Passenger.

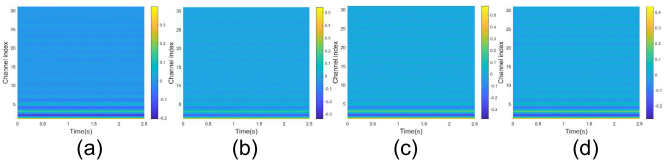


Fig. 8. GFCC characteristics of acoustic signals of different targets. (a) Tug. (b) Cargo. (c) Tanker. (d) Passenger.

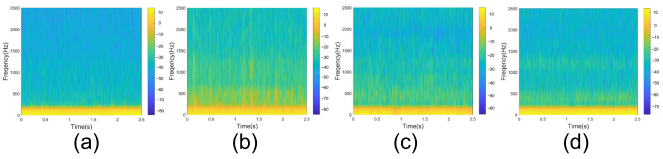


Fig. 9. LOFAR spectrum of acoustic signals of different targets. (a) Tug. (b) Cargo. (c) Tanker. (d) Passenger.

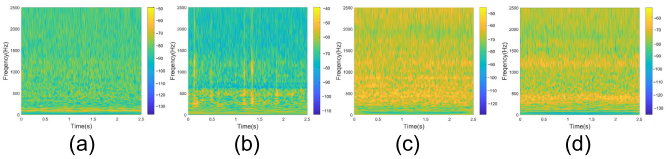


Fig. 10. CQT features of acoustic signals of different targets. (a) Tug. (b) Cargo. (c) Tanker. (d) Passenger.

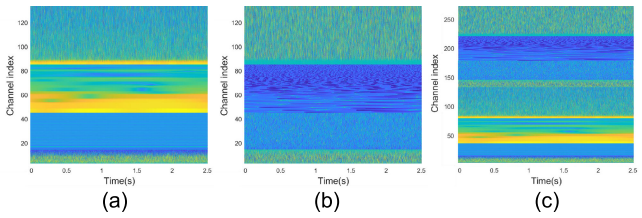


Fig. 11. Features after the first fusion and second fusion. (a) MGCL. (b) Delta. (c) MGCL-Delta.

C. Feature Fusion and Dimensionality Reduction

Because compared to matrix data samples, the CNN classifier are better at extracting rich features from the image samples, this article converts various features into color images, which are parallelized as input features for the training. The four extracted features are parallelized to obtain the first fusion feature, namely MGCL. Its dimension is 1×134. Taking 2.5 s audio data of tug as an example, the image sample of MGCL is shown in Fig. 11(a). On the basis of MGCL, its Delta feature is calculated, as shown in Fig. 11(b). It shows that the Delta feature can more clearly extract the dynamic information. MGCL and its Delta feature are combined to obtain the second fused feature with the dimension of 1×268, namely MGCL-Delta, as shown in Fig. 11(c).

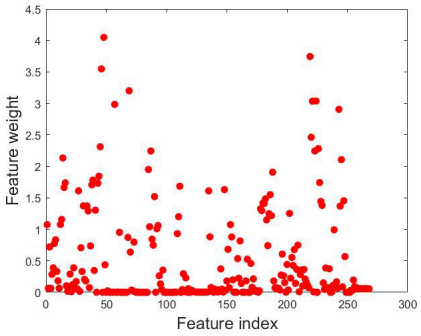


Fig. 12. NCA feature screening diagram of MGCL-Delta.

TABLE II
DIMENSIONS OF DIFFERENT FEATURES BEFORE AND AFTER DIMENSIONALITY REDUCTION BY THE NCA

Feature	MFCC	GFCC	CQT	LOFAR
Before NCA	12	31	41	50
After NCA	7	17	11	12

For feature fusion, to avoid the problem of excessively high dimension after feature parameters are mixed, this study uses NCA method to reduce feature dimensions. Fig. 12 illustrates the NCA feature screening diagram of MGCL-Delta. The recognition task requires distinguishing the signals of different ship, so the threshold feature weight should not be too high. This article sets the threshold to 0.3. Features with weights less than 0.3 can be considered as not representing the specific information of the data sample, which are removed to achieve to screen the features. Table II shows the dimensions of different features before and after dimensionality reduction by the NCA. Only MFCC, GFCC, CQT, and LOFAR spectrum are displayed here, and the dimension of their Delta features is not listed. The results show that the NCA method can effectively remove redundant information, retaining only representative features. Especially CQT and LOFAR spectrum, a large amount of redundancy has been removed. The screened MGCL-Delta features after NCA are represented as MGCL-Delta-NCA, with the dimension of 1×96.

After reducing the dimensionality of the fusion features, this article converts the four types of features and their Delta features in the MGCL-Delta-NCA into color images, which are concatenated in parallel to obtain the input features for the recognition. This article obtains the 2-D single image sample by parallelizing 1-D features of five consecutive frames. Fig. 13 shows the 2-D single input image samples of MGCL-Delta-NCA for the different targets. It can be observed that there are obvious differences in the fusion features of different targets.

Fig. 14 shows the statistical histograms of multiple characteristics. Different colors represent four kinds of target acoustic signals. The results show that the MFCC, MGCL, and MGCL-Delta of the four target acoustic signals are relatively similar, which leads to the degradation of recognition performance to a certain extent. The MGCL-Delta-NCA spectrum

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TABLE III
RECOGNITION ACCURACY OF 1-D FEATURES AND 2-D COLOR IMAGES FOR DIFFERENT FEATURES (%)

Feature	MFCC	GFCC	CQT	LOFAR	MGCL	MGCL -Delta	MGCL -Delta-NCA
One-dimensional feature	86.33	83.85	86.22	80.25	89.41	93.62	94.47
Two-dimensional color image	90.16	87.09	89.31	84.58	92.59	95.35	97.29

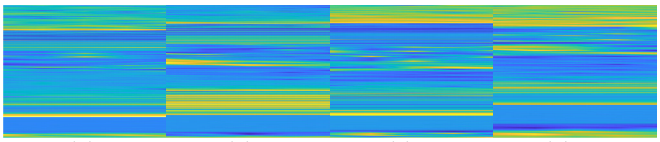


Fig. 13. Input image samples of MGCL-Delta-NCA for the different targets. (a) Tug. (b) Cargo. (c) Tanker. (d) Passenger.

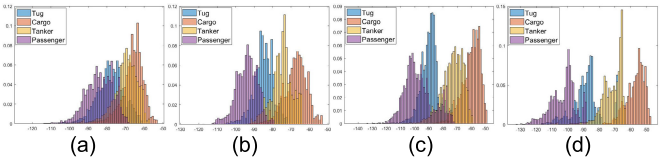


Fig. 14. Statistical histogram of different input features. (a) MFCC. (b) MGCL. (c) MGCL-Delta. (d) MGCL-Delta-NCA.

of the four kinds of target acoustic signals have evident differences, which are manifested in the kurtosis, shape and position of the distribution. Therefore, compared with other features, the feature information of the MGCL-Delta-NCA is more representative, making it more suitable as the input for the recognition tasks.

In this article, the MFCC, GFCC, CQT, and LOFAR spectrogram of 11 800 segments are extracted from each type of acoustic target. On the basis of feature fusion, 2360 2-D single image samples of MGCL-Delta-NCA are obtained by parallelizing 2-D features of five consecutive frames, with 3/4 of the image samples randomly selected for the training and the remaining 1/4 for the testing. Namely, 1770 training samples and 590 testing samples are used for each type of ship. As a result, the training set and test set contain 7080 and 2360 samples, respectively.

D. Recognition Results

The recognition results of support vector machine (SVM), FNN, ordinary CNN, GRNN, VGG19, and ResNet18 are analyzed and compared. For the SVM, radial basis function (RBF) is applied as kernel function [27]. For the FNN, it has a hidden layer, and the sigmoid and softmax functions are used to activate the hidden and output layer [28]. The ordinary CNN model includes three convolution layers, called 3-CNN. For the GRNN, it only has one network parameter, namely spread factor [29]. The k-fold cross validation method is used to determine the optimal spread factor, and k is taken as 10. As for the VGG19, it consists of 16 convolution layers, five pooling layers, and three full connection layers [30]. ResNet18 and VGG19 have 17 and 16 convolution layers, respectively, with the similar numbers of layers. The significant difference between ResNet18 and VGG19 is that adjacent convolutional layers of ResNet18 uses residual connection, while VGG19 only adds convolutional layers based on shallow CNN model. To analyze the results of different methods systematically, this article uses precision, recall, and F1 score as measurement indices, whose equations are

$$\text{Accuracy} = \frac{\text{TP} + \text{TN}}{\text{TP} + \text{TN} + \text{FP} + \text{FN}} \tag{25}$$

$$\text{precision} = \frac{\text{TP}}{\text{TP} + \text{FP}} \tag{26}$$

$$\text{recall} = \frac{\text{TP}}{\text{TP} + \text{FN}} \tag{27}$$

$$\text{F1-score} = 2 * \frac{\text{precision} * \text{recall}}{\text{precision} + \text{recall}} \tag{28}$$

where TP, TN, FP, and FN are true positive, true negative, false positive, and false negative, respectively.

For the different features, this study compares the results of 1-D features with those of color image samples obtained by parallelizing the 1-D features of several consecutive frames, which are input into the ResNet18, respectively. The recognition results of 1-D features and 2-D color images of different features are shown in Table III. The results show that the recognition performance of features can be obviously improved by parallelizing the 1-D features of several consecutive frames and drawing the color image samples. Among the MFCC, GFCC, CQT, and LOFAR spectrum, MFCC feature extraction is more in-depth and systematic, that is, feature information close to speech task has been extracted before inputting the machine-learning model. Moreover, MFCC can more effectively extract low-frequency feature information, and the feature information of ship sound is precisely shown in the low-frequency part. The ERB scale has the high resolution in the low-frequency part, and the cubic root nonlinear compression of GFCC is better than the logarithmic compression of MFCC. Nevertheless, the advantage of GFCC lies in its good noise resistance, which has not been fully utilized in noise-free scenarios. CQT features can also include much low-frequency information. However, it is applicable for the situations with long frames. In practical applications, the number of samples is generally small, and the length of frame selected in this study is relatively short, which has more research significance. The LOFAR spectrum lacks the low-frequency information, and it only processes on the original audio preliminarily. Therefore, under the same method, the accuracy of MFCC is higher than that of GFCC, LOFAR, and CQT.

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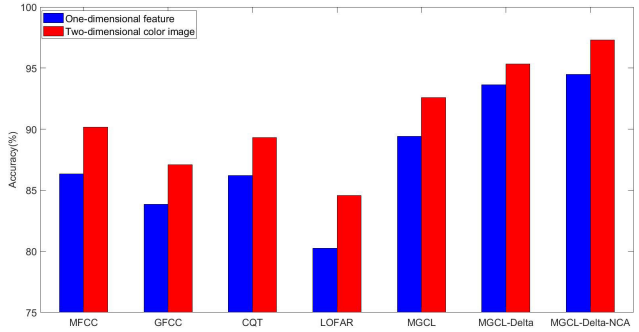


Fig. 15. Recognition results of 1-D features and color image samples obtained by parallelizing the 1-D features of several consecutive frames.

To more intuitively demonstrate the difference in recognition performance between 1-D features and 2-D color images, this article presents the recognition results of the various features in Fig. 15. It can be observed that the recognition results of 2-D color images of MGCL-Delta-NCA are the best. Due to the limited feature information provided by 1-D features and the machine-learning model being better at extracting the differences of image samples, the recognition results of the 2-D color image samples obtained by splitting the previous individual frames are significantly better than those of 1-D features. Compared to GFCC, CQT, and LOFAR, the 2-D color images of MFCC has the best recognition results. Therefore, this article only uses the 2-D color image samples of MFCC, MGCL, and MGCL-Delta for comparison with the MGCL-Delta-NCA.

The accuracy of various ships for inputting MGCL-Delta-NCA into the ResNet18 is shown in Table IV. The results show that FNN belongs to the artificial neural networks, which can extract the feature information through one-way propagation, therefore, compared with traditional SVM based on statistical theory, the recognition results of FNN are better. Due to the shallow model structure and lack of back propagation capability, FNN is unable to extract deep features, which causes significant errors due to underfitting. However, 3-CNN has more layers with the characteristics of local perception and back propagation. Therefore, the recognition results of 3-CNN are better than those of FNN. Because the optimal spread factor of GRNN can be obtained through sufficient cross validations, the trained model is particularly close to the recognition task, resulting in better recognition results than those of 3-CNN. Increasing the depth of network can improve the training performance, so compared with 3-CNN and GRNN, VGG19 with many convolutional layers have higher recognition accuracy. However, VGG19 causes certain errors due to its tendency to overfit. The ResNet18 model using residual connections not only increases the depth of the network, but also effectively avoids overfitting. Therefore, compared to the VGG19 model, ResNet18 model has better recognition results. Namely, the ResNet18 has the best recognition performance among the above models.

The recognition results of different methods of inputting MGCL-Delta-NCA, MGCL-Delta, MGCL, and MFCC are shown in Tables V–VIII, respectively. To compare the recognition performance of different features more intuitively, this article presents the recognition accuracy of multiple models for inputting different features, as shown in Fig. 16. MFCC can more effectively extract low-frequency feature information, and the feature information of ship sound is precisely shown in the low-frequency part. MGCL utilizes feature fusion technology to combine the advantages of MFCC and CQT, which have rich low-frequency feature information,

TABLE IV
TARGET SIGNAL RECOGNITION ACCURACY (%) FOR INPUTTING MGCL-DELTA-NCA

Method	A	B	C	D	Total
SVM	84.65	86.52	85.80	87.91	86.22
FNN	87.17	88.40	86.09	88.86	87.63
3-CNN	87.81	88.71	89.36	91.12	89.25
GRNN	93.41	91.15	92.34	92.98	92.47
VGG19	96.18	94.54	93.42	93.58	94.43
ResNet18	98.45	95.39	97.51	97.81	97.29

TABLE V
RECOGNITION RESULTS (%) OF VARIOUS MODELS FOR INPUTTING MGCL-DELTA-NCA

Method	Accuracy	Precision	Recall	F1 Score
SVM	86.22	86.26	86.17	86.21
FNN	87.63	87.68	87.61	87.65
3-CNN	89.25	89.33	89.21	89.27
GRNN	92.47	92.54	92.50	92.52
VGG19	94.43	94.49	94.35	94.42
ResNet18	97.29	97.34	97.25	97.29

TABLE VI
RECOGNITION RESULTS (%) OF VARIOUS MODELS FOR INPUTTING MGCL-DELTA

Method	Accuracy	Precision	Recall	F1 Score
SVM	82.77	82.81	82.74	82.78
FNN	84.28	84.30	84.22	84.26
3-CNN	86.39	86.42	86.32	86.37
GRNN	89.12	89.17	89.09	89.13
VGG19	93.03	93.09	92.98	93.04
ResNet18	95.35	95.41	95.33	95.37

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TABLE VII
RECOGNITION RESULTS (%) OF VARIOUS MODELS FOR INPUTTING MGCL

Method	Accuracy	Precision	Recall	F1 Score
SVM	78.50	78.54	78.47	78.51
FNN	81.09	81.17	81.04	81.10
3-CNN	82.85	82.93	82.82	82.88
GRNN	84.94	85.02	84.90	84.96
VGG19	91.31	91.36	91.28	91.32
ResNet18	92.59	92.63	92.53	92.58

TABLE VIII
RECOGNITION RESULTS (%) OF VARIOUS MODELS FOR INPUTTING MFCC

Method	Accuracy	Precision	Recall	F1 Score
SVM	73.21	73.27	73.13	73.20
FNN	77.33	77.36	77.26	77.31
3-CNN	80.81	80.90	80.75	80.82
GRNN	83.59	83.66	83.56	83.61
VGG19	87.52	87.60	87.47	87.54
ResNet18	90.16	90.23	90.12	90.17

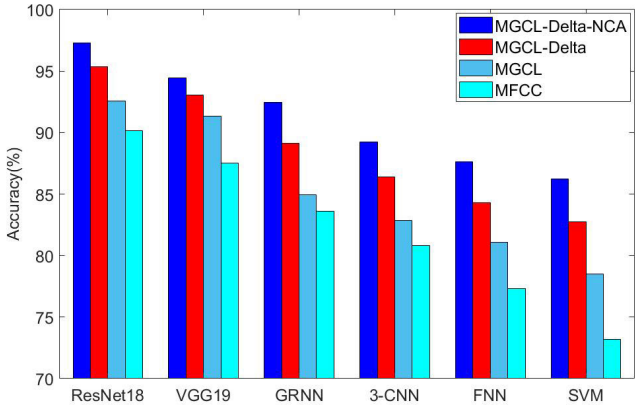


Fig. 16. Recognition accuracy of multiple models for inputting different features.

a large amount of information at different frequency points in the LOFAR spectrum, and the robustness of GFCC in various scenarios. Therefore, compared to MFCC features, MGCL has better recognition results. MGCL-Delta adds differential coefficients containing a large amount of signal dynamic information on the basis of MGCL, which improves recognition performance to a certain extent. However, it contains a large amount of redundant feature information, making it difficult for machine-learning models to extract features that are closest

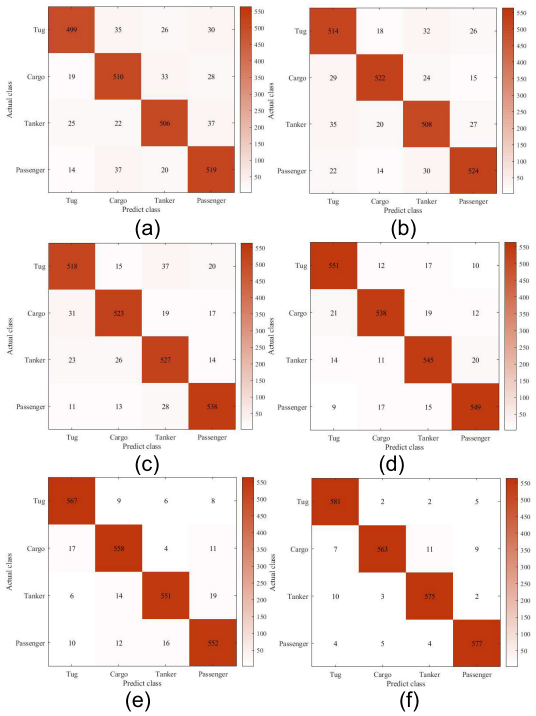


Fig. 17. Confusion matrix of results for the different methods with the input of MGCL-Delta-NCA. (a) SVM. (b) FNN. (c) 3-CNN. (d) GRNN. (e) VGG19. (f) ResNet18.

to the recognition task, thereby limiting their recognition performance. MGCL-Delta-NCA not only integrate the advantages of multiple features, but also remove redundant features by NCA, greatly improving the quality and richness of training samples. Therefore, at the same method, MGCL-Delta-NCA has the best recognition performance.

To comprehensively analyze the recognition results of multiple methods, this study draws the recognition results of MGCL-Delta-NCA into confusion matrices, as shown in Fig. 17. It shows that the recognition results of SVM, FNN, and 3-CNN for many samples are incorrect. GRNN and VGG19 can only achieve approximate recognition. Compared to other methods, ResNet18 has better test results, and most test samples can be accurately recognized.

E. At Low SNRs

In the actual environment, the sea generally has various levels of environmental noise. This study analyzes the generalization capabilities of multiple methods in different complex underwater environments by adding Gaussian white noise with various signal-to-noise ratios (SNRs) to the original acoustic signal. Take the original signal of a small ship and the waveform of the signal at different SNR as an example, as shown in Fig. 18, the fewer the SNR, the more burrs emerge in the waveforms of original signal.

This study compared the recognition performance of multiple methods in the different SNR scenarios for inputting MFCC, MGCL, MGCL-Delta, and MGCL-Delta-NCA, whose recognition results are shown in Tables IX–XIV, respectively. Due to the fact that SVM is based on traditional statistical theory, it can only recognize approximately with the input of

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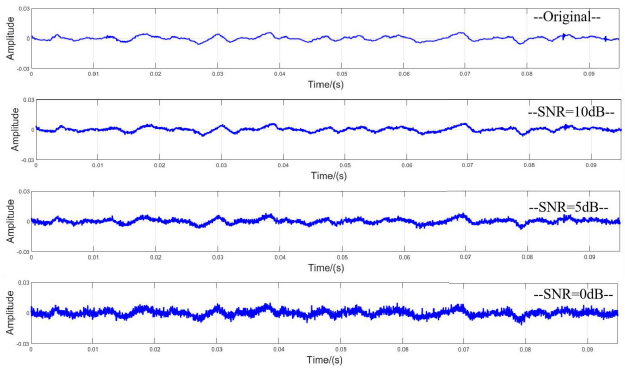


Fig. 18. Waveforms of original signal and different SNR.

TABLE IX
RECOGNITION ACCURACY (%) OF SVM AT VARIOUS SNRS

SNR	MFCC	MGCL	MGCL-Delta	MGCL-Delta-NCA
20	74.62	79.85	81.56	85.89
10	67.49	74.63	77.51	80.69
5	65.45	70.53	74.68	76.23
0	60.31	66.59	69.33	73.86

MGCL-Delta and MGCL-Delta-NCA. When inputting MFCC and MGCL, SVM is limited by the insufficient richness of input features, resulting in the inability to extract representative feature information. Therefore, it can no longer achieve effective recognition, especially at low SNRs. Compared to SVM, FNN has better nonlinear fitting and self-adaption abilities, resulting in better recognition performance. However, FNN belongs to lightweight neural network models and can only extract feature information through one-way propagation, so it can only recognize approximately at high SNRs. With the same SNR and input feature, compared to SVM and FNN, 3-CNN has better recognition results. However, due to the limitation of fewer layers, 3-CNN can only extract shallow features, so it is also not applicable at low SNRs. For inputting MGCL-Delta-NCA, VGG19 can achieve accurate recognition at high SNRs, but only rough recognition can be achieved at low SNRs. For the ResNet18, effective recognition can be achieved by inputting different features at high SNRs. However, at low SNRs, ResNet18 can still recognize robustly only with input of MGCL-Delta-NCA. Especially at 0dB, the recognition accuracy still reaches 86.25%.

The relationship curve between the recognition results of various models and SNR is shown in Fig. 19. It can be observed that MFCC is greatly affected by SNR and cannot be applied at low SNRs. MGCL and MGCL Delta can only achieve rough recognition at low SNRs. MGCL-Delta-NCA combines multiple feature advantages and removes redundancy. Therefore, with the same method and SNR, it has the best recognition results and is least affected by SNR. The recognition performance of SVM, FNN, 3-CNN, and GRNN decreases significantly due to the effect of low SNRs.

TABLE X
RECOGNITION ACCURACY (%) OF FNN AT VARIOUS SNRS

SNR	MFCC	MGCL	MGCL-Delta	MGCL-Delta-NCA
20	75.95	80.19	83.95	87.05
10	70.35	76.29	79.42	84.13
5	67.40	73.05	75.79	79.34
0	64.02	69.44	72.05	76.18

TABLE XI
RECOGNITION ACCURACY (%) OF 3-CNN AT VARIOUS SNRS

SNR	MFCC	MGCL	MGCL-Delta	MGCL-Delta-NCA
20	78.90	81.15	85.51	88.28
10	71.47	75.69	80.93	84.89
5	66.61	72.85	76.06	81.15
0	62.23	68.01	73.58	77.69

TABLE XII
RECOGNITION ACCURACY (%) OF GRNN AT VARIOUS SNRS

SNR	MFCC	MGCL	MGCL-Delta	MGCL-Delta-NCA
20	82.80	84.17	88.53	92.09
10	74.61	79.45	84.46	87.30
5	71.22	75.20	78.32	84.49
0	66.51	72.65	75.50	80.72

TABLE XIII
RECOGNITION ACCURACY (%) OF VGG19 AT VARIOUS SNRS

SNR	MFCC	MGCL	MGCL-Delta	MGCL-Delta-NCA
20	85.14	89.90	92.02	93.22
10	79.39	84.83	87.51	89.61
5	74.03	80.61	82.81	85.90
0	68.62	77.53	80.08	82.29

With the same input feature and SNR, ResNet18 has the best recognition performance. As the SNR decreases, its corresponding curve of recognition results descends most smoothly. The MGCL-Delta-NCA-ResNet18 model fully utilizes the rich feature information advantage of MGCL-Delta-NCA and capability of residual CNN to extract the deep feature, so it has the best recognition results, which can still recognize effectively at low SNRs. Namely, it has the excellent generalization capability in different complex underwater environments.

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TABLE XIV
RECOGNITION ACCURACY (%) OF RESNET18 AT VARIOUS SNRS

SNR	MFCC	MGCL	MGCL-Delta	MGCL-Delta-NCA
20	88.61	91.01	94.03	96.16
10	83.05	86.93	88.82	91.31
5	77.41	83.56	85.89	89.62
0	73.38	79.05	83.05	86.25

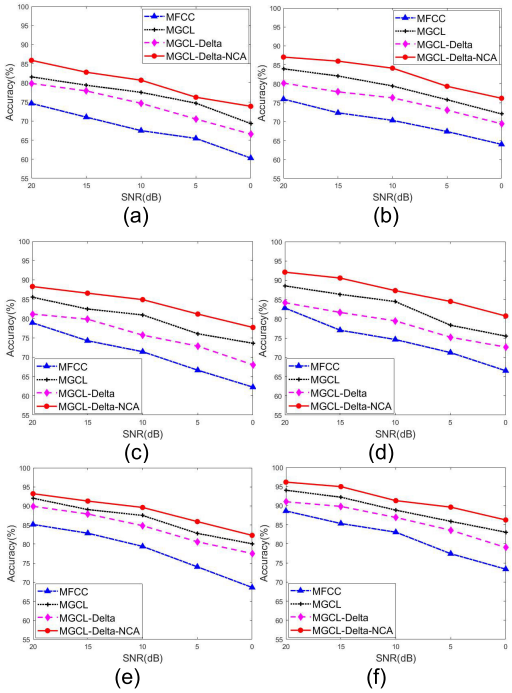


Fig. 19. Underwater acoustic target recognition results of multiple methods at different SNRs. (a) SVM. (b) FNN. (c) 3-CNN. (d) GRNN. (e) VGG19. (f) ResNet18.

F. Multiple Target Scenario

Taking dual targets and triple targets as examples, this article analyzes the recognition performance of the proposed model in the multiple target scenario. The signal of multiple targets is obtained by adding two or three signals point by point and taking the average value. Tug, cargo, and tanker are represented as A, B, and C, respectively. A + B represents the superimposed signal of the tug and cargo, and so on. Fig. 20 shows the signal waveform in the scenarios with multiple targets. The number of features obtained from different types of signals in the multiple target scenario is the same as that in the single target scenario. 3/4 of the image samples randomly are also selected for the training and the remaining 1/4 are used for the testing. Namely, 1770 training samples and 590 testing samples are used for each type of signal in the multiple target scenario. As a result, the training set and test set contain 12 390 and 4130 samples, respectively.

This article uses n-hot encoding form to generate training labels. Table XV shows the labels of training samples in scenarios with multiple targets. Where, each dimension in the vector represents whether the sample contains signals of the corresponding category, with one containing the corresponding signal, otherwise it is 0.

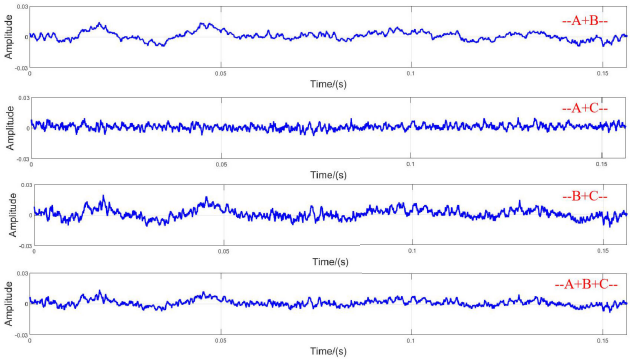


Fig. 20. Underwater acoustic target recognition results of multiple methods at different SNRs.

TABLE XV
LABELS OF TRAINING SAMPLES IN THE SCENARIOS WITH MULTIPLE TARGETS

A	B	C	A+B	A+C	B+C	A+B+C
1	0	0	1	1	0	1
0	1	0	1	0	1	1
0	0	1	0	1	1	1

TABLE XVI
TARGET SIGNAL RECOGNITION ACCURACY (%) FOR INPUTTING VARIOUS FEATURES IN THE MULTIPLE TARGET SCENARIO

Method	MFCC	MGCL	MGCL-Delta	MGCL-Delta-NCA
SVM	66.19	69.02	74.62	80.93
FNN	68.33	74.19	78.21	83.12
3-CNN	71.85	75.31	81.86	85.49
GRNN	74.88	77.40	84.17	86.54
VGG19	76.91	80.42	85.54	89.66
ResNet18	79.32	84.15	88.63	91.39

The recognition results for inputting different features in the multiple target scenario are shown in Table XVI. The recognition performance of different features in the multiple target scenario is compared more intuitively, as shown in Fig. 21. The results show that in the multiple target scenario, due to the increase in the required number of classifications and the fact that multitarget signals contain some feature information similar to the single target, the complexity of the recognition task is relatively high. It is no longer applicable with MFCC and MGCL. For inputting MGCL-Delta and MGCL-Delta-NCA, both VGG19 and ResNet18 can achieve effective recognition. However, 3-CNN and GRNN can only

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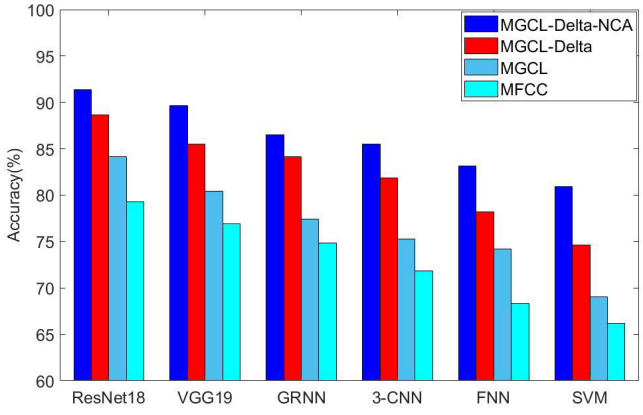


Fig. 21. Recognition accuracy for inputting various features in the multiple target scenario.

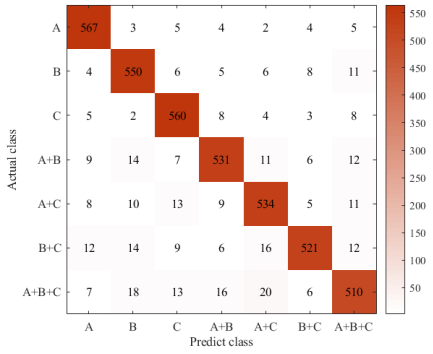


Fig. 22. Confusion matrices of ResNet18 for inputting MGCL-Delta-NCA.

recognize approximately. The MGCL-Delta-NCA-ResNet18 model has the best recognition results and can still achieve recognition accuracy of over 90%.

The confusion matrices of ResNet18 for inputting MGCL-Delta-NCA is shown in Fig. 22. The results show that the MGCL-Delta-NCA-ResNet18 model can recognize multitarget signal effectively, which meets the accuracy requirements of practical applications. Due to the possibility that the model may only extract the feature information of partial targets for the multiple target signals, compared to single target signals, the recognition results of multiple target signals has more errors.

G. Comparison With the State-of-the-Art Models and Fusion Strategies

Table XVII shows the comparison of the recognition accuracy with different state-of-the-art models. Where, GoogLeNet [31], DenseNet-121 [32], and MobileNet-V3 [33] are representative state-of-the-art machine-learning models, with the input of MGCL-Delta-NCA as an example. Mel-Transformer [34], MFCC-2D-CNN [35], MFCC-CDBN [14], and LOFAR-AMNet [7] are the applications of machine learning in the field of underwater acoustic recognition, which have shown the good recognition performance. The results show that although GoogLeNet and DenseNet-121 have deep network architectures, their recognition results are limited by overfitting, gradient dispersion, and only a few underwater acoustic samples. MobileNet-V3 is composed of the separable convolution and linear bottleneck inverse residual structure, with the excellent ability to extract feature. Therefore, compared to GoogLeNet and DenseNet-121, the recognition results are better. However, MobileNet-V3 belongs to lightweight neural network models, which is not particularly suitable for the recognition tasks in the complex scenes. The feature information of LOFAR spectrum is relatively preliminary with much redundancy. Moreover, as classifier, LENET is limited by the relatively shallow depth, which cannot extract deep features. Therefore, the recognition results of LOFAR-LENET are not good enough. Mel-spectrogram only represents the frequency spectrum of the signal on the Mel-frequency scale, lacking dynamic feature information. Although transformer adopts attention architecture to extract the global dependency relationship of sample data, it is weak in obtaining local feature information. However, the differences of underwater targets are generally reflected in the local and detailed features. Therefore, the application of Mel-Transformer model in underwater target recognition has certain limitations. MFCC-2D-CNN contains multiple convolutional layers without the residual connections, making it prone to overfitting. Moreover, the retention of the pooling layer leads to the filtration of some useful features. The network architecture and parameters of MFCC-CDBN rely heavily on empirical information. The inappropriate parameter can lead to the model converging to local optimal solution. The convolutional attention module are introduced in the LOFAR-AMNet, which greatly improves the recognition performance. However, its recognition performance is still limited in the single type of acoustic feature. For the above models, most of them use the traditional features of MFCC, LOFAR, and Mel-spectrogram. Compared to LOFAR and Mel-spectrogram, MFCC has more abundant low-frequency features, which are closer to the recognition task. However, only MGCL-Delta-NCA achieve the multiangle representation of target signals through the feature fusion technology. The proposed MGCL-Delta-NCA-ResNet18 model not only systematically integrates multiple acoustic features and the corresponding Delta features, but also removes a large amount of redundancy by the NCA method. Especially, the deep CNN with residual connections, namely ResNet18, is used as a classifier to effectively extract deep features. Therefore, due to the aforementioned factors, MGCL-Delta-NCA-ResNet18 has the best recognition results.

Table XVIII shows the comparison of the recognition accuracy with different fusion strategies. Where, APB-INN represents the integrated neural network (INN) to fuse amplitude spectrum, phase spectrum, and bispectrum feature [36]. STFT, Mel-spectrogram, and MFCC are integrated by the embedding fusion of convolution recurrent neural network (CRNN), called SMM-CRNN [37]. Time domain data and Mel-spectrogram are fused to input the CNN with self-attention module, called TM-CNN-SA [38]. Long short-term memory (LSTM) network is used to fuse time domain data, LOFAR and Mel-spectrogram, called TLM-LSTM [39]. The proposed method fuses MFCC, GFCC, CQT, LOFAR, and their corresponding Delta features in parallel, and then

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TABLE XVII
COMPARISON OF THE RECOGNITION ACCURACY (%) WITH DIFFERENT STATE-OF-THE-ART MODELS

Model	GoogLeNet	DenseNet -121	MobileNet -V3	LOFAR -LENET	Mel -Transformer	MFCC -2D-CNN	MFCC -CDBN	LOFAR -AMNet	MGCL-Delta -NCA-ResNet18
Accuracy	91.35	89.28	94.53	89.19	93.65	92.12	90.49	94.05	97.29

TABLE XVIII
COMPARISON OF THE RECOGNITION ACCURACY (%) WITH DIFFERENT FUSION STRATEGIES

Model	APB -INN	SMM -CRNN	TM -CNN-SA	TLM -LSTM	MGCL-Delta -Superposition	MGCL-Delta -Stack	MGCL-Delta -NCA-Parallel	MGCL -Delta-NCA
Accuracy	92.25	94.39	92.77	87.52	85.16	92.31	95.02	97.29

uses NCA to perform the reduction of feature dimension, namely MGCL-Delta-NCA. MGCL-Delta-NCA-Parallel represents first reducing the dimensions of above four types of acoustic features and their Delta features, and then fusing them in parallel. MGCL-Delta-Superposition represents adding all the 2-D feature matrices of MGCL-Delta. MGCL-Delta-Stack represents inputting all the 2-D feature matrices of MGCL-Delta into different channels of the model. For the MGCL-Delta-Superposition and MGCL-Delta-Stack, the zero padding operation is used to ensure consistency in the dimensions of different 2-D feature matrices. The classifiers for MGCL-Delta-Parallel-NCA, MGCL-Delta-NCA-Parallel, MGCL-Delta-Superposition, and MGCL-Delta-Stack are all ResNet18. The results show that APB-INN, TM-CNN-SA, and TLM-LSTM significantly increase the complexity of the model, which easily lead to overfitting and gradient explosion. The decision-making processes for these models between different sub-networks are also difficult, resulting in limited recognition performance. Although the SMM-CRNN model combines CNN and RNN to fuse features, it encounters many configuration and optimization problems of hyperparameters, and excessive dropout layers may also filter out much useful feature information. In addition, the above fusion strategies only integrate a small number of features and does not extract their Delta features. Especially, the time domain data, amplitude spectrum, phase spectrum, Mel-spectrogram, and bispectrum are relatively preliminary and lack abundant low-frequency feature information. MGCL-Delta-Superposition greatly reduces the richness of features and cannot fully utilize the advantages of multiple features. The excessive zero padding operations and lack of feature dimensionality reduction in the MGCL-Delta-Stack result in significant redundance, which reduces recognition performance. Performing NCA after feature fusion is beneficial for more accurately removing a large amount of redundancy, making the obtained features closer to the recognition task. Therefore, compared to MGCL-Delta-NCA-Parallel, MGCL-Delta-Parallel-NCA has better recognition results. MGCL-Delta-Parallel-NCA not only fuses the most types of features, but also avoids the redundancy problem caused by excessive feature parameters. Especially, the overall framework is not too complex and does not require configuring and optimizing many hyperparameters. Therefore, compared with other fusion strategies mentioned above, MGCL-Delta-NCA-Parallel has better recognition performance and greater practical value.

H. Computational Load

This article analyzes the computational load of different models and features. The models are developed on a workstation with 11th Gen Intel[1] Core[2] i7-1165G7 CPU*8. Taking the input of MGCL-Delta-NCA as an example, Table XIX shows the training and testing time of different models. With ResNet18 as the classifier, Table XX shows the computing, training, and testing time for inputting different features. Where, the computing time represents the time required for the feature extraction and fusion. The results show that due to the shallow depth of SVM, FNN, GRNN, and 3-CNN, both training and testing time are relatively short. GoogLeNet, DenseNet-121, and MobileNet-V3 have complex structures with too many layers, so their training time is relatively long. The residual connection of the ResNet18 model skips some convolutional layers, which can promote the convergence and reduce the computational load. Therefore, the training time of ResNet18 is less than that of VGG19. Compared to only inputting the single acoustic feature, using feature fusion technology has more computing, training, and testing time. However, it can significantly improve recognition performance. Although compared to other fusion strategies, MGCL-Delta-Superposition has the shorter training and testing time, its recognition performance decreases significantly. MGCL-Delta-Stack not only has the most feature parameters, but also greatly increases the number of channels, resulting in the longest training and testing time. MGCL-Delta-NCA-Parallel requires performing NCA on all the features before the feature fusion, so its computing time is the longest. The training and testing time of MGCL-Delta-NCA is shorter than that of MGCL-Delta-Stack and MGCL-Delta-NCA-Parallel. Besides, the difference in the testing time for different features is small. In conclusion, although the training time of MGCL-Delta-NCA-ResNet18 is relatively not too short, it can achieve more accurate recognition. Moreover, for the trained MGCL-Delta-NCA-ResNet18 model, it effectively recognizes the test samples in a short time. Therefore, it can be used for real-time recognition of

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TABLE XIX
TRAINING AND TESTING TIME OF DIFFERENT METHODS FOR INPUTTING MGCL-DELTA-NCA

Method	SVM	FNN	3-CNN	GRNN	VGG19	GoogLeNet	DenseNet -121	MobileNet -V3	ResNet18
Training time/s	25.3	33.8	42.6	36.9	85.1	98.2	92.7	74.3	70.8
Testing time/s	1.4	1.9	2.1	1.6	2.9	3.6	3.3	2.5	2.3

TABLE XX
COMPUTING, TRAINING, AND TESTING TIME OF DIFFERENT FEATURES

Method	MFCC	GFCC	CQT	LOFAR	MGCL-Delta -Superposition	MGCL-Delta -Stack	MGCL-Delta -NCA-Parallel	MGCL -Delta-NCA
Computing time/s	2.0	2.3	1.5	1.1	7.4	7.2	9.7	8.5
Training time/s	10.2	11.8	8.9	14.9	42.3	91.5	73.8	70.8
Testing time/s	0.8	0.9	0.6	1.3	1.6	4.9	2.7	2.3

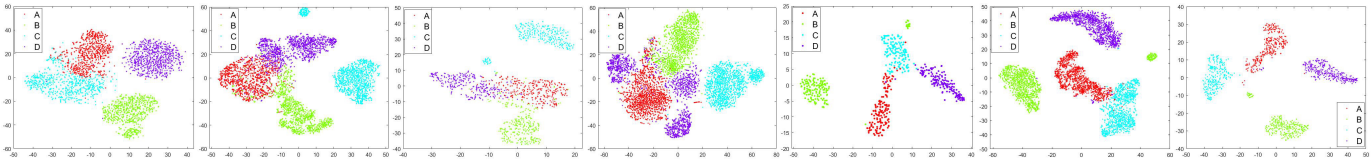


Fig. 23. t-SNE visualization results of the fully connected layer using different features. (a) MFCC. (b) GFCC. (c) CQT. (d) LOFAR. (e) MGCL. (f) MGCL-Delta. (g) MGCL-Delta-NCA.

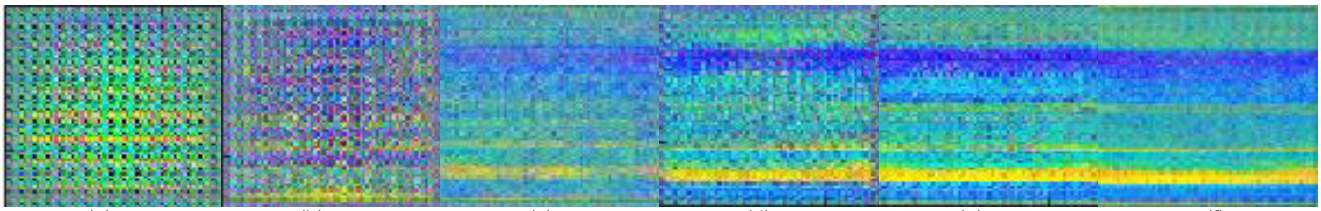


Fig. 24. Visualization results of different layers for the MGCL-Delta-NCA-ResNet18. (a) Conv1. (b) Conv2. (c) Conv5. (d) Conv9. (e) Conv14. (f) FC.

underwater acoustic targets and meets the accuracy requirements of practical applications.

I. Visualization

With ResNet18 as the classifier, this article analyzes and compares the t-distributed stochastic neighbor embedding (t-SNE) visualization results of different features, as shown in Fig. 23. The output features of the fully connected layer are processed using t-SNE to obtain the corresponding positions in the planar graph [40]. Different colors similarly represent four types of acoustic target signals. The closer the distance between each type of acoustic target signal, the greater the similarity. The results show that compared to the single acoustic feature, the distance between different types of acoustic target samples using feature fusion technology is obviously larger. Besides, the samples of the same type are more compact. Therefore, the feature fusion technology can greatly improve the recognition performance. Especially, the distance between different types of targets using MGCL-Delta-NCA is the largest. Namely, compared to other features, MGCL-Delta-NCA is more suitable for recognition tasks, resulting in better recognition performance.

With the input of MGCL-Delta-NCA as an example, this article also analyzes the visualization results of different layers of ResNet18, as shown in Fig. 24. Where, conv-n represents nth convolutional layer, and FC represents the fully connected layer. The results show that the deeper the network layer, the more detailed the generated filters, which can be trained to obtain more complex patterns and textures. Moreover, MGCL-Delta-NCA-ResNet18 can clearly fit the input features of acoustic target signal, mainly reflected in color and texture, so that the trained model can effectively recognize underwater acoustic targets.

J. Summary of Recognition Results

From the recognition results, it can be obtained that MGCL-Delta-NCA-ResNet18 can achieve effective identification in different scenarios, and compared to other models,

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its recognition results are better. Its specific innovations and advantages are mainly reflected in the following aspects. First, for the specific feature fusion strategy, this article performs the feature fusion of MFCC, GFCC, CQT, and LOFAR in parallel. The corresponding Delta features are also calculated and fused, which particularly contain abundant dynamic information. Namely, MGCL-Delta can systematically reflect the target sound signal from multiple angles. Second, this article uses NCA for the reduction of acoustic feature dimension with the supervised way. It can be trained to obtain projection matrices through continuous iterative optimization, which greatly removes redundancy to improve recognition performance and reduces the computational resources required for the machine-learning model training. Besides, the used ResNet18 model with residual connections not only increases the depth of the network, but also effectively avoids overfitting. Therefore, it has the ability to extract abundant deep feature information. Finally, the proposed method systematically combines the extraction of multiple acoustic features, including dynamic Delta features, multiangle fusion feature technology, feature dimensionality reduction of NCA, and the ability of ResNet18 to extract deep feature, fully utilizing their respective advantages. Therefore, it can accurately recognize targets in the multiple scenarios with the excellent performance.

In the actual marine environment, the widespread presence of noise, the scenario of multiple target signals, and the scarcity of data samples greatly increase the complexity of the recognition task, which poses the significant challenges. For the proposed MGCL-Delta-NCA-ResNet18, compared with the state-of-the-art models and fusion strategies, it has better recognition performance due to the systematic utilization of advantages of multiple technologies. Moreover, the feature fusion technology designed in this article can comprehensively improve the richness and noise resistance. Therefore, in the scenarios with low SNRs and multiple targets, it can still achieve the effective target recognition. Especially, due to the relatively low computational resources, it can be used for real-time recognition of underwater acoustic targets with the great practical value. Therefore, based on the above factors, the MGCL-Delta-NCA-ResNet18 has high accuracy, real-time capability, and robustness in the multiple complex scenarios with only limited sample data, making it particularly suitable for the field of underwater acoustic target recognition.

VII. Conclusion

Aiming at the common problems, such as insufficient effective dimension of underwater acoustic feature parameters and poor anti-noise performance, MFCC, GFCC, CQT, and LOFAR spectrum are extracted and fused first. Their Delta features are calculated and fused second. The feature dimension is reduced by NCA. Then, the fusion feature parameters are inputted into ResNet18 for the recognition. The results show that MFCC features have poor recognition results at low SNRs, whereas GFCC features have better anti-noise performance. Because MGCL-Delta-NCA-ResNet18 allows full play to the rich information advantages of feature fusion, advantages of feature dimensionality reduction by the NCA and the ability of ResNet18 to extract abundant characteristics, among these models, MGCL-Delta-NCA-ResNet18 has best recognition results, with the recognition accuracy of 97.29%. At low SNRs, it can also recognize effectively. Particularly at 0 dB, the recognition accuracy can still reach 86.25%. The proposed method can also recognize multitarget signal effectively in the multiple target scenario. While this method is used in ship and natural voice recognition, it can be applied to other target voice recognition, such as marine mammals. In theory, due to the multiangle and abundant feature information of feature fusion technology and the excellent feature extraction ability of ResNet18, the proposed method can be further extended to the recognition of unseen targets. However, this poses a challenge to whether the model can maintain high accuracy, which needs further research and exploration.

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Yixin Yang (Member, IEEE) received the B.S. degree in applied electronic engineering and the M.S. and Ph.D. degrees in underwater acoustic engineering from Northwestern Polytechnical University (NPU), Xi'an, China, in 1997, 1999, and 2002, respectively.

From June 2002 to June 2004, he was a Research Fellow with the School of Electrical and Electronic Engineering, Nanyang Technological University, Singapore. Since July 2004, he has been with the School of Marine Science and Technology, NPU, where he became a Professor in 2006. He is currently the Vice President of NPU. His research interests include underwater acoustics, sonar array signal processing, signal detection and parameter estimation, and target recognition.

Dr. Yang is the Chair of the Underwater Acoustics Committee of the Acoustical Society of China.



Qihai Yao was born in 1997. He received the master's degree in ship and ocean engineering from Northwestern Polytechnical University, Xi'an, China, in 2022, where he is currently pursuing the Ph.D. degree in ship and ocean engineering.

In the first half of 2020, he completed the exchange project of the Oxford Prospects program of the University of Oxford, U.K., and obtained an excellent graduation certificate. His main research direction is the application of machine learning in target recognition and array signal processing.



Yong Wang (Member, IEEE) was born in 1987. He received the B.S. degree in applied electronic engineering and the M.S. and Ph.D. degrees in underwater acoustic engineering from Northwestern Polytechnical University (NPU), Xi'an, China, in 2009, 2011, and 2015, respectively.

From 2016 to 2017, he was a Postdoctoral Fellow with Xi'an Jiaotong University, Xi'an, China. He is currently an Associate Professor with the School of Marine Science and Technology, NPU. His research interests include array signal processing, parameter estimation, and sonar signal processing.

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